DSAI 352 — Computer Vision

Spring 2026

Bonus Project Specification

On-Device Mobile Application for RC-Car Bowling Score Detection

Authorship and status. This document was compiled by Amr Yasser, a student in the course, from notes taken during the instructor's video briefing of the bonus project. The contents were subsequently confirmed in person with the instructor, Dr Khaled, who verified the requirements and the grading structure described herein. The teaching assistant is not involved in grading this bonus project.

1. Course and Project Information

Course Code	DSAI 352
Course Title	Computer Vision
Term	Spring 2026
Deliverable	Bonus Project (separate from the main course project)
Discussion Date	Wednesday, 6 May 2026, during the regular Computer Vision lecture slot (12:00 PM – 2:00 PM)
Group Size	Individual. Each student is graded alone, regardless of who collaborated on recordings.
Registration	Closed. Registration was opened in a single lecture in which students wrote their name and ID, or had them written on their behalf. Registration is no longer accepted.
Grader	The instructor (Dr Khaled), not the teaching assistant. The instructor conducts the discussion and assigns the grade personally.

2. Grading and Risk Structure

This bonus project carries an asymmetric grading structure that the team should understand before committing further effort.

The bonus is added on top of the student's existing grade in the course, capped so that the total grade for the course cannot exceed 60 out of 60. The bonus does not affect the maximum grades of the final exam, the coursework component, or the midterm individually.

Outcome	Effect on grade
Project completed correctly and successfully defended in the discussion.	+5 points
Project completed correctly, successfully defended, and includes the optional path-detection feature (Section 5.4).	+7 points
Registered student fails to submit, submits work that is not their own, or cannot answer the instructor's questions during the discussion.	−3 points

The instructor framed the grading philosophy in terms of professional accountability: a company that commits to a customer deliverable and fails risks both its reputation and its insurance liability. Registration for this bonus project is treated as the analogous commitment, and failure to deliver carries a corresponding penalty.

Students who did not register may still present their work to the instructor on a courtesy basis and receive feedback, but cannot receive any bonus grade.

3. Project Overview

The bonus project is a variation of the main course project. The same bowling-with-pins setup is used, but with two substantive differences.

The first difference is the platform. The complete pipeline — video capture, on-device inference, scoring, and rendering — must run on a mobile device, fully offline. The application may target either Android or iOS at the student's discretion. Sending the video to a backend server, calling a remote API, or otherwise depending on a network connection at inference time is explicitly disallowed.

The second difference is the means by which the pins are knocked down. Instead of a thrown bowling ball, a small remote-controlled toy car is used. The student drives the car using a handheld remote, manoeuvring it (forwards, backwards, left, right) to strike the pins. As with the main project, the recording and the driving are split between two people: one student records on the mobile application, and the other operates the remote. A shared physical car and shared pins may be used across all students; this is permitted.

The mobile application produces an annotated output video. The output displays a numerical label on each pin in the order it fell (the first pin to fall is labelled 1, the second is labelled 2, and so on). At the end of the recording, the application displays the total elapsed time and the total number of pins knocked down during that period.

4. Live Demonstration Requirement

The work is not submitted as a pre-recorded video alone. The instructor will hold a discussion session at the regular Computer Vision lecture slot on Wednesday 6 May 2026 (12:00 PM – 2:00 PM). During this session the student must place pins in front of the instructor, drive the car to strike them, and have the mobile application record and process the run live. The instructor reviews the resulting annotated output on the mobile device.

Submitting only a pre-prepared video, including any video sourced from the internet, does not satisfy this requirement. The application is tested live; no exception is offered.

During the discussion the student must explain the techniques, model, and preprocessing used in the application, and answer the instructor's questions about the implementation.

5. Functional Requirements

5.1 Mobile application

- Records video using the device camera.
- Performs all preprocessing, inference, and rendering on-device, without any network call at inference time.
- Produces an annotated output video viewable within the application.

5.2 Pin order detection

- Detects the moment each pin falls during the recording.
- Assigns to each fallen pin a numerical label corresponding to the order in which it fell. The first pin to fall is labelled 1, the second 2, and so on.
- Renders the numerical label on the corresponding pin in the output video.

5.3 Final score summary

- Displays the total elapsed time of the run.
- Displays the total number of pins knocked down.

5.4 Path detection (optional, +2 bonus)

Optionally, the application may detect and visualise the trajectory of the car throughout the recording. The trajectory is rendered as a coloured line (the colour and persistence are at the student's discretion: the path may remain visible for the full duration of the recording, appear briefly as the car moves, or use any other reasonable visual treatment). Successful and defensible implementation of this feature raises the bonus from five points to seven.

Path detection requires single-object tracking of a small, fast-moving target across a handheld mobile recording. It is significantly more demanding than pin-order detection alone and should not be attempted as a last-minute addition. If the path-detection implementation does not work or cannot be defended in the discussion, only the base bonus (five points) is at stake — not the negative penalty — provided the core pin-order task is complete.

6. Constraints and Restrictions

- The application must run fully offline at inference time. Network calls during inference are not permitted.
- Target platform: Android or iOS, at the student's discretion.
- The video used in the demonstration must be captured live by the student's own application during the discussion session. Pre-recorded videos and any video downloaded from the internet are not accepted.
- Number of pins: at least five.
- The physical equipment (remote-controlled car and pins) is shared among students. Students do not need to source their own car or pins.
- The instructor's questions during the discussion form a substantive part of the grade. Inability to answer questions about the implementation is treated as evidence that the work is not the student's own and triggers the negative penalty.

7. Modeling Expectation

Consistent with the modeling rule for the main course project, the student must build their own model, fine-tune a pretrained model, or apply transfer learning. A fully off-the-shelf, ready-made model used without any training, fine-tuning, or transfer step does not satisfy the requirement.

Within that constraint, the choice of model architecture and framework is open. The instructor confirmed that any model may be used, provided it can be made to run correctly on the mobile application. The student is responsible for selecting a programming language and framework whose models can be deployed on the chosen mobile platform, and for performing whatever conversion or packaging is required to run the trained model on-device.

8. Deliverables

The deliverables for the bonus project are as follows.

- The working mobile application, presented and tested live in front of the instructor during the discussion session.
- A written report describing the techniques used, the model and preprocessing pipeline, what was attempted, what worked, and any difficulties encountered.

Students may bring supporting materials (notebooks, training scripts, conversion logs) to help answer the instructor’s questions. The live demonstration on the mobile device, together with the written report and the discussion, constitute the artefacts under evaluation.

— End of specification —